FLOODGUARD
Comprehensive OLS MLR Model Guide

Raw evidence to final candidates | Sto. Nino, Nangka, and Tumana

Station	Final candidate	Predictors	Decision status
Sto. Nino	C9	Sto WL(t-1), Sto WL(t-3), Boso-Boso rain(t-1)	CERTIFIED_PASS
Nangka	C4	Nangka WL(t-1), Boso-Boso rain(t-1)	MODEL_SELECTION_PASS
Tumana	C8	Tumana WL(t-1), Science Garden rain(t-1)	TUMANA_MLR_TIME_SERIES_MODEL_PASS

Final position: the certified operational models remain C9, C4, and C8. Critical-day audits are retained as transparent post-selection sensitivity evidence; they do not silently rewrite the frozen original decisions.

This guide is designed for team study and panel defense. It explains what was done, why each test exists, what every major number means, what alternatives performed better on particular metrics, and why the final operational choices remain defensible.

1. Executive answer: are the models done?

Yes for the station-level OLS MLR research packages and supporting audits. The three final equations, candidate tournaments, chronological validation, HC3/VIF diagnostics, persistence comparisons, formula evidence, runtime parity tests, and post-selection critical-day audits are available. Website/backend integration and final workspace cleanup are separate engineering tasks.

Question	Short answer
What algorithm is used?	Transparent Ordinary Least Squares Multiple Linear Regression with past-day predictors.
Is it a time-series method?	Yes. Every final equation uses at least one prior water-level lag.
Did one metric decide everything?	No. Model form, calendar integrity, HC3, VIF, chronological validation, exact-common-date comparisons, persistence, availability, and event sensitivity were separated.
Are the models best on every typhoon day?	No universal claim is made. The guide reports exact trade-offs.
Why not replace models after the event audit?	The new audit was performed after the original selections and uses already-examined dates. Replacing winners would be hindsight selection without a new prospective test.

The panel-ready one-sentence explanation

We selected transparent station-specific OLS MLR time-series models through a traceable pipeline: preserve raw data, build exact calendar lags, freeze candidates, screen reliability with HC3 and VIF, validate on later dates, compare only identical dates, benchmark against persistence, then disclose availability and critical-day trade-offs.

2. What the main terms mean

Term	Simple meaning	Why it matters
OLS	Finds coefficients that minimize squared prediction errors.	Produces a visible equation that can be checked in Python, JavaScript, and Excel.
MLR	A regression with at least two predictors.	All deployed station models meet the capstone requirement.
Time-series lag	A past value such as water level at t-1.	Prevents using future information.
HC3	A cautious standard-error calculation.	Checks whether a slope still appears dependable when error spread is uneven or influential days exist. It does not change the OLS equation.
VIF	A redundancy check among predictors.	VIF <= 5 avoids unstable coefficients caused by strongly overlapping inputs.
MAE	Average absolute error.	Easy to interpret; treats each miss proportionally.
RMSE	Square-root of average squared error.	Punishes large misses more strongly; useful for surge sensitivity.
Persistence	Tomorrow equals yesterday.	A strong honest baseline for slowly changing rivers.
Exact common dates	Only dates both models predicted.	Stops a model from looking better merely because it skipped difficult days.

3. Common research pipeline: raw file to final prediction

Step	What was done
1. Source inventory	Identify station, timestamp, units, coverage, file type, and source semantics. Hash and inventory source files.
2. Parsing and quarantine	Parse telemetry values, exclude spreadsheet summaries, quarantine impossible or unparseable tokens, and record decisions.
3. Continuous calendar	Create a complete 2016-01-01 to 2025-12-31 date grid so missing dates stay visible.
4. Daily target	Sto. Nino and Nangka use their certified daily station series derived under the project completeness rules. Tumana preserves its one source-reported daily value without relabeling it as maximum or mean.
5. Exact calendar lags	Create t-1, t-2, t-3 by date matching, never by previous nonblank row.
6. Missing-data discipline	Keep missing values blank/null. No zero fill, interpolation, forward-fill, or backward-fill.
7. Freeze candidates	Write candidate predictor definitions before reading validation performance.
8. Fit training OLS	Estimate coefficients on older dates only.
9. Screen reliability	Calculate HC3 robust inference and VIF. Keep eligibility separate from predictive ranking.
10. Chronological validation	Test frozen models on later dates; never random-split the time series.
11. Fair tournament	Use candidate-specific complete cases and exact-common-date pairwise comparisons.
12. Benchmarks and operations	Compare with true persistence, report availability, refit through 2023, verify Excel/runtime parity, and perform retrospective/event audits.

Common chronological splits

Phase	Dates	Use
Training	2016-01-01 to 2022-05-26	Estimate candidates and run HC3/VIF screens.
Validation	2022-05-27 to 2023-12-31	Rank frozen candidates on later dates.
Final refit	2016-01-01 to 2023-12-31	Estimate production coefficients after the candidate is selected.
2024-2025 retrospective	2024-01-01 to 2025-12-31	Operational availability and historical sensitivity; already examined, not a new untouched holdout.

4. Source coverage and target definitions

Station	Calendar rows	Valid target values	Target wording / semantics
Sto. Nino	3,653	3,361	Certified Sto. Nino daily station water-level series under the corrected 2016-2025 pipeline.
Nangka	3,653	2,798	Certified Nangka daily station water-level series under the corrected 2016-2025 pipeline.
Tumana	3,653	2,233	PAGASA-reported daily Tumana water-level observation; source does not establish maximum, mean, or fixed-time semantics.

Why Tumana is described differently

Tumana has one supplied value per calendar day, but its metadata does not confirm what that value represents within the day. The value was preserved exactly. No hourly observations were created; no daily maximum or mean was inferred; and the output is daily decision support only, not an intraday flood-peak or evacuation trigger.

Candidate-specific complete cases

A candidate is fitted and evaluated only on dates where its own target and predictors are present. This preserves more valid evidence than forcing every model onto the smallest common dataset. For a head-to-head claim, however, the two candidates are re-compared on their exact shared dates.

Key distinction: candidate-specific samples are used to estimate and report availability; exact-common-date samples are used to declare which of two models performed better.

5. Sto. Nino: complete candidate tournament

Sto. Nino: frozen candidate definitions

ID	Model	Predictors	K
C01	Baseline AR1	Sto_t_1	1
C02	AR2 Model (AR1 + AR3)	Sto_t_1 + Sto_t_3	2
C03	WL Multi-Station (AR1 + Nangka_t_3)	Sto_t_1 + NangkaWL_t_3	2
C04	AR1 + Boso-Boso_t_1	Sto_t_1 + BosoBoso_t_1	2
C05	Candidate 5 (AR1 + Campana_t_1)	Sto_t_1 + Campana_t_1	2
C06	AR1 + NangkaRain_t_1	Sto_t_1 + NangkaRain_t_1	2
C07	AR1 + Oro_t_1	Sto_t_1 + Oro_t_1	2
C08	AR1 + Aries_t_1	Sto_t_1 + Aries_t_1	2
C09	AR2 + Boso-Boso_t_1	Sto_t_1 + Sto_t_3 + BosoBoso_t_1	3
C10	Candidate 10 (AR2 + Oro_t_1)	Sto_t_1 + Sto_t_3 + Oro_t_1	3
C11	Candidate 11 (AR2 + Campana_t_1)	Sto_t_1 + Sto_t_3 + Campana_t_1	3
C12	AR2 + Aries_t_1	Sto_t_1 + Sto_t_3 + Aries_t_1	3
C13	AR2 + Nangka_WL_t_3	Sto_t_1 + Sto_t_3 + NangkaWL_t_3	3
C14	Multi-Rain (Oro + Boso-Boso)	Sto_t_1 + Sto_t_3 + Oro_t_1 + BosoBoso_t_1	4
C15	Candidate 15 (Full Watershed: Oro + Campana)	Sto_t_1 + Sto_t_3 + Oro_t_1 + Campana_t_1	4

Evidence: Certified R2 COMPLETE_CANDIDATE_RESULTS.csv.

Sto. Nino: complete validation results

ID	Train N	Val N	HC3 slopes	Max VIF	MAE	RMSE	Pair rank
C01	2063	504	PASS	1.00	0.3147	0.5842	10
C02	1976	473	PASS	1.49	0.3093	0.5925	5
C03	1856	363	PASS	1.20	0.3137	0.6148	11
C04	2056	503	PASS	1.40	0.3230	0.5847	15
C05	1820	502	PASS	1.35	0.3087	0.5807	7
C06	2000	466	PASS	1.46	0.3190	0.5919	13
C07	2020	478	PASS	1.49	0.3213	0.5911	12
C08	2048	487	PASS	1.50	0.3139	0.5748	14
C09	1969	472	PASS	2.06	0.3163	0.5911	9
C10	1933	449	PASS	2.18	0.3085	0.5899	4
C11	1752	471	PASS	2.27	0.3033	0.5772	3
C12	1962	458	PASS	2.24	0.3078	0.5814	8
C13	1826	356	PASS	3.94	0.2920	0.6130	1
C14	1933	449	PASS	2.31	0.3178	0.5939	6
C15	1717	448	FAIL	2.73	0.3067	0.5848	2

6. Why Sto. Nino Candidate 9 was retained

Candidate 13 won the overall validation tournament with 14 pairwise wins. Candidate 9 ranked ninth with 6 wins. C9 was therefore not selected because it was the overall validation champion. It was retained as the operational prototype because it passed the final statistical checks, uses a practical dependency set, achieved 90.60% retrospective availability, and performed strongly against C15 and persistence on the package-defined critical subsets.

Final Sto. Nino equation

Sto_WL(t) = 3.545852 + 0.464200 x Sto_WL(t-1) + 0.245693 x Sto_WL(t-3) + 0.011525 x BosoBoso_Rain(t-1)

Term	Coefficient	HC3 SE	HC3 p	Interpretation
Intercept	3.545852	0.484802	3.50e-13	Fitted constant; not a standalone physical stage.
Sto WL(t-1)	0.464200	0.067102	5.84e-12	Yesterday carries river memory.
Sto WL(t-3)	0.245693	0.048432	4.21e-7	Three-day lag adds slower memory.
Boso-Boso rain(t-1)	0.011525	0.002203	1.82e-7	Prior upstream rainfall adds rain-response information.

C9 maximum VIF is approximately 2.06, below the ceiling of 5. All production slopes are HC3-significant.

Validation and retrospective evidence

Evidence lens	C9	Comparator	Honest conclusion
Validation	N=472; MAE 0.3163; RMSE 0.5911	C13 ranked first overall	C9 is not the validation winner.
Overall retrospective, exact N=557	MAE 0.2415; RMSE 0.3794	Persistence MAE 0.1643; RMSE 0.3901	Persistence wins MAE; C9 slightly lowers RMSE.
Named typhoon windows, N=24	MAE 0.7266; RMSE 1.0161	C15 MAE 0.7654; persistence 0.8367	C9 wins the examined comparison.
Operational-critical, N=30	MAE 0.9755; RMSE 1.1717	C15 MAE 1.1168; persistence 1.3610	C9 wins the examined comparison.

New exact-common critical sensitivity check

Using the cross-river sensitivity definition (training 90th-percentile high stage or rise >=0.50 m), C9 and C14 were compared on the same 62 retrospective dates. C9 had MAE 0.7215 m and RMSE 0.9154 m; C14 had MAE 0.7672 m and RMSE 0.9457 m. C9 therefore remained better on the same dates. The earlier candidate-specific table made C14 look better because it skipped three difficult C9 dates.

Known failure that must be disclosed

On 2025-07-21 during Crising, actual Sto. Nino stage was 18.60 m after a +4.80 m rise. C9 predicted about 13.7947 m, an error of about 4.81 m. This proves the model cannot guarantee sudden extreme peaks.

7. Nangka: complete candidate tournament

Nangka: frozen candidate definitions

ID	Model	Predictors	K
C01	Baseline AR1	Nangka_WL_t_1	1
C02	AR2 Model (AR1 + AR3)	Nangka_WL_t_1 + Nangka_WL_t_3	2
C03	WL Multi-Station (AR1 + Sto_Nino_t_3)	Nangka_WL_t_1 + Sto_Nino_t_3	2
C04	AR1 + Boso-Boso_t_1	Nangka_WL_t_1 + BosoBoso_t_1	2
C05	AR1 + Campana_t_1	Nangka_WL_t_1 + Campana_t_1	2
C06	AR1 + NangkaRain_t_1	Nangka_WL_t_1 + Nangka_Rain_t_1	2
C07	AR1 + Oro_t_1	Nangka_WL_t_1 + Oro_t_1	2
C08	AR1 + Aries_t_1	Nangka_WL_t_1 + Aries_t_1	2
C09	AR2 + Boso-Boso_t_1	Nangka_WL_t_1 + Nangka_WL_t_3 + BosoBoso_t_1	3
C10	AR2 + Oro_t_1	Nangka_WL_t_1 + Nangka_WL_t_3 + Oro_t_1	3
C11	AR2 + Campana_t_1	Nangka_WL_t_1 + Nangka_WL_t_3 + Campana_t_1	3
C12	AR2 + Aries_t_1	Nangka_WL_t_1 + Nangka_WL_t_3 + Aries_t_1	3
C13	AR2 + Sto_Nino_WL_t_3	Nangka_WL_t_1 + Nangka_WL_t_3 + Sto_Nino_t_3	3
C14	Multi-Rain (Oro + Boso-Boso)	Nangka_WL_t_1 + Nangka_WL_t_3 + Oro_t_1 + BosoBoso_t_1	4
C15	Full Watershed (Oro + Campana)	Nangka_WL_t_1 + Nangka_WL_t_3 + Oro_t_1 + Campana_t_1	4

Evidence: Nangka R5 NANGKA_COMPLETE_CANDIDATE_RESULTS.csv.

Nangka: complete validation results

ID	Train N	Val N	HC3 slopes	Max VIF	MAE	RMSE	Pair rank
C01	1820	323	PASS	1.00	0.1538	0.4105	12
C02	1698	257	FAIL	1.18	0.1517	0.4137	9
C03	1731	300	PASS	1.19	0.1576	0.4209	11
C04	1810	318	PASS	1.60	0.1438	0.3846	1
C05	1612	317	PASS	1.53	0.1537	0.3809	7
C06	1797	297	PASS	1.65	0.1538	0.4196	2
C07	1780	302	PASS	1.68	0.1505	0.3957	4
C08	1806	313	PASS	1.81	0.1517	0.4088	6
C09	1688	254	FAIL	1.92	0.1478	0.3933	3
C10	1660	244	FAIL	1.93	0.1535	0.4056	5
C11	1518	253	PASS	1.84	0.1544	0.3811	10
C12	1685	250	PASS	2.16	0.1538	0.4161	8
C13	1661	248	FAIL	3.06	0.1566	0.4173	13
C14	1660	244	PASS	2.23	0.1624	0.3974	14
C15	1493	243	PASS	2.18	0.1643	0.3932	15

8. Why Nangka Candidate 4 was retained

C4 passed HC3 and VIF and won the frozen overall 15-candidate validation tournament: rank 1 and 14 of 14 exact-common-date pairwise MAE wins. It was selected for overall validation generalization, not because it has the smallest RMSE or wins every event subset.

Final Nangka equation

Nangka_WL(t) = 8.114817 + 0.489769 x Nangka_WL(t-1) + 0.009737 x BosoBoso_Rain(t-1)

Evidence	C4 result	Trade-off
Validation sample	N=318; MAE 0.143768; RMSE 0.384644; R2 0.510505	Pairwise MAE rank 1; C5 has slightly lower RMSE 0.380929.
True persistence on same 318 dates	MAE 0.126774; RMSE 0.440142	Persistence wins MAE; C4 lowers RMSE by 12.61%.
Diagnostics	Max VIF 1.5957; final rain HC3 p about 1.93e-6	Passes reliability screens.
Primary availability, 2024-2025	354/424 observed target days = 83.49%	Missing rain activates true-persistence fallback when prior WL exists.

New critical-day all-eligible comparison

Model	Exact common N	Critical MAE	Critical RMSE	Interpretation
C15	16	0.9415	1.2623	Lowest MAE.
C11	16	0.9590	1.2510	Lowest RMSE.
C5	16	0.9714	1.2681	Better than C4 on both metrics.
C14	16	0.9823	1.3024	Better than C4 on both metrics.
C4 selected	16	1.0447	1.3425	Not the critical-subset winner; retains much stronger overall tournament standing and broader availability.

No single challenger unanimously wins critical MAE, critical RMSE, overall validation, and availability. C15 wins critical MAE but has only 243 validation predictions versus C4's 318 and ranks last in the original pairwise tournament. C11 wins critical RMSE but ranks tenth overall. Replacing C4 after seeing these subsets would be post-hoc re-selection.

9. Tumana: complete candidate tournament

Tumana: frozen candidate definitions

ID	Model	Predictors	K
C01	Baseline AR1	Tumana_WL_t_1	1
C02	AR2 Model (AR1 + AR3)	Tumana_WL_t_1 + Tumana_WL_t_3	2
C03	AR1 + Boso-Boso_t_1	Tumana_WL_t_1 + BosoBoso_t_1	2
C04	AR1 + Campana_t_1	Tumana_WL_t_1 + Campana_t_1	2
C05	AR1 + Oro_t_1	Tumana_WL_t_1 + Oro_t_1	2
C06	AR1 + Aries_t_1	Tumana_WL_t_1 + Aries_t_1	2
C07	AR1 + NangkaRain_t_1	Tumana_WL_t_1 + NangkaRain_t_1	2
C08	AR1 + PAGASA_SG_Rain_t_1	Tumana_WL_t_1 + PAGASA_SG_Rain_t_1	2
C09	AR1 + PAGASA_SG_RH_t_1	Tumana_WL_t_1 + PAGASA_SG_RH_t_1	2
C10	AR1 + PAGASA_SG_TAVG_t_1	Tumana_WL_t_1 + PAGASA_SG_TAVG_t_1	2
C11	AR1 + PAGASA_SG_Wind_t_1	Tumana_WL_t_1 + PAGASA_SG_Wind_t_1	2
C12	AR1 + PAGASA_SG_HeatIndex_t_1	Tumana_WL_t_1 + PAGASA_SG_HeatIndex_t_1	2
C13	AR2 + Boso-Boso_t_1	Tumana_WL_t_1 + Tumana_WL_t_3 + BosoBoso_t_1	3
C14	AR2 + PAGASA_SG_Rain_t_1	Tumana_WL_t_1 + Tumana_WL_t_3 + PAGASA_SG_Rain_t_1	3
C15	Regional Multi-Rain (Oro + BosoBoso)	Tumana_WL_t_1 + Tumana_WL_t_3 + Oro_t_1 + BosoBoso_t_1	4

Evidence: Tumana R3.1.2 TUMANA_COMPLETE_CANDIDATE_RESULTS.csv.

Tumana: complete validation results

ID	Train N	Val N	HC3 slopes	Max VIF	MAE	RMSE	Pair rank
C01	1068	580	PASS	1.00	0.1442	0.3543	3
C02	1043	578	FAIL	2.58	0.1463	0.3488	11
C03	1061	547	FAIL	1.19	0.1413	0.3492	5
C04	1056	546	FAIL	1.19	0.1394	0.3449	4
C05	1027	520	FAIL	1.26	0.1455	0.3602	10
C06	1055	531	FAIL	1.23	0.1450	0.3600	9
C07	1016	495	FAIL	1.23	0.1487	0.3707	15
C08	1068	580	PASS	1.12	0.1481	0.3244	1
C09	1068	580	FAIL	1.21	0.1471	0.3525	13
C10	1068	580	FAIL	1.01	0.1478	0.3529	14
C11	1068	580	FAIL	1.03	0.1433	0.3505	7
C12	1068	573	FAIL	1.00	0.1468	0.3555	12
C13	1037	545	FAIL	3.46	0.1419	0.3277	6
C14	1043	578	PASS	3.01	0.1514	0.3101	2
C15	1004	518	FAIL	4.00	0.1440	0.3267	8

10. Why Tumana Candidate 8 was retained

MLR is only the minimum model form, not the reason C8 won. Among statistically eligible MLR candidates, C8 passed HC3 and VIF and won the frozen primary rule: lower validation MAE and the eligible-only exact-common-date pairwise MAE comparison against C14. C14's lower RMSE is disclosed.

Final Tumana equation

Tumana_WL(t) = 1.514724 + 0.873544 x Tumana_WL(t-1) + 0.008482 x PAGASA_SG_Rain(t-1)

Term	Coefficient	HC3 SE	HC3 p	Meaning
Intercept	1.514724	0.768255	0.0488	Fitted constant.
Tumana WL(t-1)	0.873544	0.062491	4.83e-42	Strong prior-day memory.
Science Garden rain(t-1)	0.008482	0.001481	1.22e-8	Added prior-day rainfall response.

Maximum VIF is 1.12. Both slopes pass HC3. The target is strictly the PAGASA-reported daily Tumana water-level observation.

C8 versus C14 and persistence

Comparison	C8	Other	Conclusion
Overall validation	MAE 0.1481; RMSE 0.3244	C14 MAE 0.1514; RMSE 0.3101	C8 wins frozen MAE; C14 wins RMSE.
Persistence, same N=580	MAE 0.1481; RMSE 0.3244	Persistence MAE 0.1416; RMSE 0.3604	Persistence wins MAE; C8 lowers RMSE by 9.99%.
Critical union, same N=44	MAE 0.7023; RMSE 0.9960	C14 MAE 0.6506; RMSE 0.9140	C14 wins post-selection critical sensitivity.
2024-2025 availability	C8 primary: 311/566 = 54.95%	With fallback: 547/566 = 96.64%; 19 unavailable	Rain ends after 2024; Jan 1, 2025 can still use Dec 31 rain.

C14 is a credible future challenger, but the critical audit was added after the original selection. Without a new prospective evaluation, changing C8 would rewrite the selection rule after seeing the answer.

11. Cross-river comparison of the retained final models

Dimension	Sto. Nino C9	Nangka C4	Tumana C8
MLR/time-series	K=3; WL t-1, WL t-3, rain t-1	K=2; WL t-1, rain t-1	K=2; WL t-1, rain t-1
Original selection basis	Operational balance; C13 won validation	Overall tournament rank 1, 14/14 wins	Eligible-MLR MAE and pairwise win vs C14
HC3/VIF	All final slopes significant; VIF about 2.06	Rain significant; VIF 1.60	Both slopes significant; VIF 1.12
Persistence trade-off	Persistence lower overall retrospective MAE	Persistence lower validation MAE; C4 lower RMSE	Persistence lower MAE; C8 lower RMSE
Critical audit	C9 beats C14 on exact N=62	C15 MAE winner; C11 RMSE winner on N=16	C14 beats C8 on exact N=44
Final decision	Keep C9	Keep C4	Keep C8 pending prospective challenger test

Why retaining the original models is scientifically safer

The original candidates were selected under frozen rules. The new high-stage/rapid-rise analysis was performed afterward and uses already-examined validation or retrospective dates. It is valuable as a stress test, but it is not an untouched second selection dataset. Keeping the original decisions while reporting the challengers avoids hindsight selection and preserves reproducibility.

The correct claim is not “our model is best at every flood.” The correct claim is “our final model is the most defensible operational choice under its frozen selection process, and we openly report where other candidates perform better.”

12. Excel, code, and reproducibility proof

Proof layer	Sto. Nino	Nangka	Tumana
Formula workbook	Certified defense and raw-to-daily workbooks; recalculation QA	R5 formula proof; 5,415 live formulas; 0 errors	R3.1.2 formula proof; 7,167 live formulas; 0 errors
Runtime parity	Certified implementation evidence	Python and Node agree within 1e-15	Python and Node agree within 1e-15
Manifest/checksum	Certified R2 ZIP SHA-256 4f125ab0...7857dd4d	R5 ZIP SHA-256 d787621a...d9648f8	R3.1.2 ZIP SHA-256 b84441b1...85e38cb
Missing inputs	No fabricated values; explicit unavailable/fallback rules	Rain missing -> persistence; prior WL missing -> unavailable	Rain missing -> persistence; prior WL missing -> unavailable

Why the typhoon workbook cannot replace the Sto. Nino package

The separate FLOODGUARD_TYPHOON_MAJOR_EVENT_PERFORMANCE_PROOF.xlsx evaluates exploratory Candidate 5/Candidate 15 switching and explicitly labels the switch post-hoc. It is useful historical event evidence, but it is not the certified C9 equation and must not be cited as if it proves C9 uses Oro or Campana.

13. Limitations that should be stated before the panel

Limitation	Honest statement
Daily horizon	The models forecast one calendar day ahead; they are not sub-daily hydrodynamic peak models.
Extreme underprediction	Large sudden rises can exceed what daily linear memory and rainfall lags can anticipate.
Telemetry gaps	No imputation is used. Missing prior water level can make forecasts unavailable.
Persistence strength	Quiet rivers make yesterday's level a difficult benchmark to beat on MAE.
Post-selection event audits	They reveal trade-offs but cannot be presented as a new untouched holdout.
Tumana semantics	The supplied daily value is not confirmed as daily maximum, mean, or fixed-time reading.
Safety governance	Forecasts support monitoring; live observations and official warning protocols control emergency decisions.

Claims to avoid

Unsafe claim	Correct replacement
“The model predicts every typhoon peak.”	“The model supports next-day station-level monitoring and has documented critical-day errors.”
“HC3 improves the forecast.”	“HC3 strengthens inference about slope reliability; it does not alter the OLS prediction.”
“Candidate 1 is persistence.”	“Candidate 1 is a fitted AR1 regression; persistence repeats the prior observation exactly.”
“C4 has the lowest RMSE among all Nangka candidates.”	“C4 won the frozen pairwise MAE tournament; C5 has slightly lower validation RMSE.”
“C8 is best on dangerous Tumana days.”	“C8 won the original eligible-MLR MAE rule; C14 won the post-selection critical sensitivity audit.”

14. Panel questions and defensible answers

Question	Answer
Why OLS MLR?	It is transparent, auditable, explainable, and meets the capstone requirement while incorporating river memory and rainfall.
Why chronological validation?	A real forecast learns from the past and predicts later dates. Random mixing would leak future patterns.
Why exact calendar lags?	Previous-row lags could jump across missing days and create false timing. Date matching prevents that.
Why HC3?	Storm and calm-day errors may have different spread. HC3 gives a more cautious uncertainty estimate without changing the equation.
Why VIF <= 5?	It prevents redundant predictors from making coefficients unstable and difficult to interpret.
Why keep persistence?	It is a strong benchmark and safe deterministic fallback when optional rainfall is unavailable.
Why not change to every event winner?	The event audit was conducted after selection and uses already-examined dates. Switching would be hindsight selection without prospective confirmation.
Does Tumana forecast a daily maximum?	No. It forecasts the PAGASA-reported daily Tumana water-level observation because source semantics are unconfirmed.
What is the final answer?	Sto. Nino C9, Nangka C4, and Tumana C8 remain the official operational candidates; all critical-day challengers and limitations are disclosed.

15. Evidence map: where every claim comes from

Station / topic	Authoritative evidence
Sto. Nino core	Certified R2 ZIP; COMPLETE_CANDIDATE_RESULTS.csv; CANDIDATE9_FINAL_REFIT_COEFFICIENTS_HC3.csv; validation predictions; all-candidate retrospective predictions; pairwise/event files; formula workbooks.
Nangka core	R5 reconciled ZIP; complete candidate results; ranking; persistence comparisons; exact-common pairwise file; final-refit HC3/VIF; event and availability files; formula workbook; deployment parity test.
Tumana core	R3.1.2 final evidence ZIP; target definition; complete candidate results; eligibility decision matrix; persistence comparison; final refit; availability audit; formula workbook; parity test.
Critical sensitivity	Separate R1 audit ZIPs for Sto. Nino, Nangka, and Tumana plus the cross-river exact-common-date matrix.
Workspace cleanup	Only after a corrected read-only audit identifies true package identities by hash, predictors, equation, and dataset version. Rejected candidate evidence remains required.

Final research conclusion

FloodGuard now has three transparent OLS MLR time-series model packages with auditable source handling, frozen candidates, reliability diagnostics, chronological validation, persistence benchmarks, deployment rules, formula evidence, and critical-day sensitivity disclosures. The retained final operational candidates are Sto. Nino C9, Nangka C4, and Tumana C8.

End of comprehensive guide. This document is a reconciled explanation of the evidence packages; it does not alter raw data, coefficients, candidate definitions, predictions, or certified package checksums.